Periodic Trends

Atomic radius, ionization energy, electronegativity and ionic radius

Name _____ Class _____ Date _____

DATA

First ionization energy, period 2 (kJ/mol): Li 520.2, Be 899.5, B 800.6, C 1086.5, N 1402.3, O 1313.9, F 1681.0, Ne 2080.7

First ionization energy, group 1 (kJ/mol): Li 520.2, Na 495.8, K 418.8, Rb 403.0, Cs 375.7

Electronegativity (Pauling): Na 0.93, Cl 3.16, Br 2.96, I 2.66

$Z_{\text{eff}} \approx Z - S$, where S = number of inner (core) electrons

A. Understanding

1. Define effective nuclear charge. Use $Z_{\text{eff}} \approx Z - S$ to estimate it for the outer electrons of (a) sodium (Z = 11) and (b) sulfur (Z = 16).

2. Explain why atomic radius decreases from left to right across a period.

B. Ranking

3. Arrange in order of increasing atomic radius: Mg, Ca, S, Na.

4. Arrange in order of increasing first ionization energy: K, Li, Na, Ne.

© 2026 Chemistry Clarity · chemistryclarity.com · Free for personal and classroom use. Not for resale.

5. Arrange in order of increasing electronegativity: Cl, I, Na, Br.

C. Ionization energy

6. Describe and explain the trend in first ionization energy down group 1, using the data above.

7. Across period 2, first ionization energy generally increases, but B is lower than Be and O is lower than N. Explain both exceptions.

D. Ionic radius

8. (a) Which is larger, Cl or Cl^-? (b) Which is larger, Mg or Mg^{2+}? Explain each briefly.

E. Challenge

9. S^{2-}, Cl^-, K^+ and Ca^{2+} all have 18 electrons. Arrange them in order of decreasing radius and explain your order.

© 2026 Chemistry Clarity · chemistryclarity.com · Free for personal and classroom use. Not for resale.